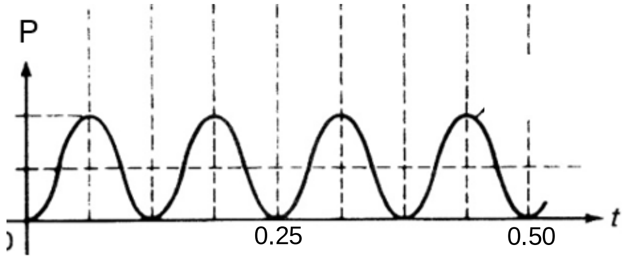
4 (a)	Root-mean-square value of an alternating current is defined as the value of the steady direct current which would dissipate heat at the same average heating effect /rate in a given resistance as the alternating current.	[1]
(b)(i)	For e.m.f. to be maximum, $\sin(\omega t) = 1$. Hence, max e.m.f. $E_o = NBA(2\pi f)$ $= (800)(0.50)(5.0\times10^{-2} \times 8.0\times10^{-2})(2\pi)(240/60)$ $= 40.2$ V	[1] [1]
(ii)	$I_o = \frac{V_o}{R} = \frac{40.2}{0.6+11.4} = 3.35\ A$	[1]
(iii)	$I_{rms} = \frac{I_o}{\sqrt{2}} = \frac{3.35}{\sqrt{2}}$ = 2.37 A	[1]
(iv)	 Label $P_{max} = I_o^2 R = (3.35^2)(11.4) =$ <u>128</u> W Label period correctly Correct shape with 2 cycles drawn.	[1] [1] [1]